

Database Programming - Homework #1

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Q1) SQL Operations

1. Total number of orders placed by each customer

```
SELECT customer_id, COUNT(order_id) AS total_orders
FROM Orders
GROUP BY customer_id;
```

2. Customers with more orders than the average

```
SELECT customer_id, COUNT(order_id) AS total_orders
FROM Orders
GROUP BY customer_id
HAVING COUNT(order_id) > (
    SELECT AVG(order_count)
    FROM (
        SELECT COUNT(order_id) AS order_count
        FROM Orders
        GROUP BY customer_id
    ) AS SubQuery
);
```

3. Creating an index on the customer_id column

```
CREATE INDEX idx_customer_id ON Orders(customer_id);
```

4. Transaction for inserting a new order and items

```
BEGIN TRANSACTION;
BEGIN TRY
    -- Insert a new order record
    INSERT INTO Orders (order_id, customer_id, order_date, total_amount)
    VALUES (201, 1, '2026-04-13', 250.00);

    -- Insert the corresponding order item
    INSERT INTO Order_Items (item_id, order_id, product_name, quantity, price)
    VALUES (601, 201, 'Monitor', 2, 125.00);

    COMMIT TRANSACTION;
END TRY
BEGIN CATCH
```

```
ROLLBACK TRANSACTION;  
END CATCH;
```

5. Explanation of ACID Properties

- **Atomicity:** The transaction is an all-or-nothing unit.
- **Consistency:** Maintains database integrity rules after the transaction.
- **Isolation:** Transactions do not interfere with each other.
- **Durability:** Committed data is permanent and survives failures.

Q2) SQL Window Functions

Cumulative total sum for quantity

```
SELECT
    day,
    qty,
    SUM(qty) OVER (ORDER BY day) AS cumQty
FROM demand;
```

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